Midterm Exam I

Statistics 1101

- ⋆ Write your name on each item.
- ⋆ Answer the multiple choice questions on the provided sheet.
- ⋆ Answer the short answer questions in your blue book.
- ⋆ You may detach the stapled pages.
- ⋆ Turn in this exam booklet with your blue book.
- ⋆ You may not use a calculator.
- ⋆ This a closed book, closed notes exam.
- ⋆ You don't have to reduce fractions or simplify square roots.
  - ✔ $\sqrt{\frac{10}{2}}$ and $\frac{1}{\sqrt{2}}$ are acceptable.
  - ✖ $\frac{2+0}{1\times 2}$ should be simplified to $\frac{2}{2}$ or 1.

Academic Integrity Agreement:
I understand what resources are and are not allowed on this exam. I won't cheat, discuss this exam with students who haven't yet taken the exam, or otherwise violate the university academic integrity standards. I will report any observed misconduct to the instructor or the appropriate authority as per university reporting procedures.

☐ I agree to the above. 👍

Name:

Circle your answers on this page. Answer short answer questions in your blue book.

Select your exam version: ♡ ♢ ♣ ♠

1.)	A	B
2.)	A	B
3.)	A	B
4.)	A	B
5.)	A	B
6.)	A	B
7.)	A	B
8.)	A	B
9.)	A	B
10.)	A	B
11.)	A	B
12.)	A	B
13.)	A	B
14.)	A	B
15.)	A	B
16.)	A	B

17.)	A	B	C	D
18.)	A	B	C	D
19.)	A	B	C	D
20.)	A	B	C	D
21.)	A	B	C	D
22.)	A	B	C	D
23.)	A	B	C	D
24.)	A	B	C	D
25.)	A	B	C	D
26.)	A	B	C	D
27.)	A	B	C	D
28.)	A	B	C	D

Cumulative z-table (below)

z	+0.00	+0.01	+0.02	+0.03	+0.04	+0.05	+0.06	+0.07	+0.08	+0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633

Interior z-table (below)

z	+0.00	+0.01	+0.02	+0.03	+0.04	+0.05	+0.06	+0.07	+0.08	+0.09
0.0	0.0000	0.0080	0.0160	0.0239	0.0319	0.0399	0.0478	0.0558	0.0638	0.0717
0.1	0.0797	0.0876	0.0955	0.1034	0.1113	0.1192	0.1271	0.1350	0.1428	0.1507
0.2	0.1585	0.1663	0.1741	0.1819	0.1897	0.1974	0.2051	0.2128	0.2205	0.2282
0.3	0.2358	0.2434	0.2510	0.2586	0.2661	0.2737	0.2812	0.2886	0.2961	0.3035
0.4	0.3108	0.3182	0.3255	0.3328	0.3401	0.3473	0.3545	0.3616	0.3688	0.3759
0.5	0.3829	0.3899	0.3969	0.4039	0.4108	0.4177	0.4245	0.4313	0.4381	0.4448
0.6	0.4515	0.4581	0.4647	0.4713	0.4778	0.4843	0.4907	0.4971	0.5035	0.5098
0.7	0.5161	0.5223	0.5285	0.5346	0.5407	0.5467	0.5527	0.5587	0.5646	0.5705
0.8	0.5763	0.5821	0.5878	0.5935	0.5991	0.6047	0.6102	0.6157	0.6211	0.6265
0.9	0.6319	0.6372	0.6424	0.6476	0.6528	0.6579	0.6629	0.6680	0.6729	0.6778
1.0	0.6827	0.6875	0.6923	0.6970	0.7017	0.7063	0.7109	0.7154	0.7199	0.7243
1.1	0.7287	0.7330	0.7373	0.7415	0.7457	0.7499	0.7540	0.7580	0.7620	0.7660
1.2	0.7699	0.7737	0.7775	0.7813	0.7850	0.7887	0.7923	0.7959	0.7995	0.8029
1.3	0.8064	0.8098	0.8132	0.8165	0.8198	0.8230	0.8262	0.8293	0.8324	0.8355
1.4	0.8385	0.8415	0.8444	0.8473	0.8501	0.8529	0.8557	0.8584	0.8611	0.8638
1.5	0.8664	0.8690	0.8715	0.8740	0.8764	0.8789	0.8812	0.8836	0.8859	0.8882
1.6	0.8904	0.8926	0.8948	0.8969	0.8990	0.9011	0.9031	0.9051	0.9070	0.9090
1.7	0.9109	0.9127	0.9146	0.9164	0.9181	0.9199	0.9216	0.9233	0.9249	0.9265

Experiments

Experiment: The researcher knows and controls the assignment of treatment or control.
Observational Study: Treatment and control groups are either not determined or not understood by the researcher.
Natural Experiment: An observational study where the assignment of treatment and control is arguably random or exogenous.

Contemporaneous Control Group: A group of subjects that is observed simultaneously with the treatment group of a study.
Historical Control Group: A control group that was observed prior to the treatment group of a study.

Descriptive Statistics

$$x_1 \leq \cdots \overbrace{x_{\frac{n+1}{2}}}^{\text{median}} \cdots \leq x_n \quad \text{if } n \text{ is odd}$$

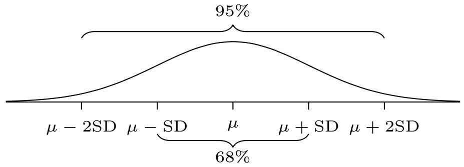
$$x_1 \leq \cdots \overbrace{x_{\frac{n}{2}}, x_{\frac{n}{2}+1}}^{\text{median}=\frac{x_{\frac{n}{2}}+x_{\frac{n}{2}+1}}{2}} \cdots \leq x_n \quad \text{if } n \text{ is even}$$

$$\text{average} = \mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\text{rms size} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2} \qquad \text{SD} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2}$$

The area of a block in a **histogram** corresponds to the percentage of data in that class interval. The height of a block represents crowding.

Normal curve



$$\text{Standardized Value} = \frac{x - \text{average}}{\text{SD}}$$

Correlation and Regression

$$r = \frac{1}{n}\sum_{i=1}^{n}\frac{(x-\mu_x)(y-\mu_y)}{\text{SD}_x \times \text{SD}_y}$$

$\hat{y} = mx + b$, where $m = r\frac{\text{SD}_y}{\text{SD}_x}$ and $b = \mu_y - m\mu_x$.

$$\text{residual} = \text{actual } y \text{ - predicted } y$$

$$\text{rms error} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\text{residual}_i^2} = \sqrt{1-r^2} \times \text{SD}_y$$

68% of y values in the interval $\hat{y} \pm$ rmse and 95% of the y values are in the interval $\hat{y} \pm 2$rmse if the scatter diagram is homoscedastic and the residuals follow a normal curve.

Probability

Conditional Probability:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \textit{ and } B)}{\mathbb{P}(B)}$$

General Multiplication Rule:

$$\mathbb{P}(A \textit{ and } B) = \mathbb{P}(A \mid B)\mathbb{P}(B)$$

Independence: A and B are independent if

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Draws from a box **with replacement** are independent.
When draws are made **without replacement**, the draws are dependent. A ticket that is drawn from the box is no longer available for the next draw, changing the probabilities of drawing the remaining tickets.

You have exam version ♣.

Following the instructions on the previous page counts for two points.

Multiple Choice (select one)

Two-choice questions are worth two points each. Four-choice questions are worth four points each. Select the best answer. Questions may continue onto the next page.

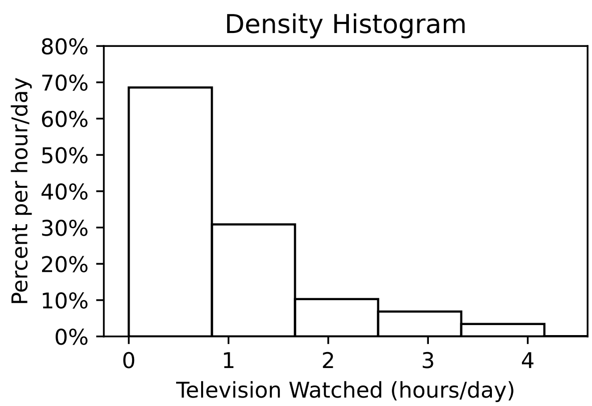


Figure 1

1. Which of these is true based on the distribution in Figure 1?
   - A. The average is greater than the median.
   - B. The median is greater than the average.
2. The SD of 3, 5 is
   - A. 1.
   - B. $\sqrt{2}$.
3. The SD of 3, 4, 5 is
   - A. SD ≥ 1.
   - B. SD < 1.
4. Correlation does not imply causation because a correlation coefficient is unitless.
   - A. True
   - B. False
5. Correlation never implies causation because x and y values are paired non-randomly.
   - A. True
   - B. False

6. A school offers an optional study hall for students. The principal of this school wants to use statistics to investigate whether this study hall is beneficial for students. The principal takes attendance at the study hall and then compares the GPA of those who attended to those who didn't. This study is

   A. Observational

   B. Experimental

7. If the correlation coefficient between two variables is zero, then there can still be a causal relationship between the two variables.

   A. True

   B. False

8. Which of these is better described as two independent things? Assume Mayor Adams and Professor Springs both live in the same city but they don't interact.

   A. Mayor Adams wears snow boots on a particular day and Professor Springs wears snow boots on the same day.

   B. Mayor Adams owns green boots and Professor Springs owns purple boots.

Use the following information for the next two questions.

The Roman Catholic Church is led by a Pope. The Papal seat becomes vacant upon the death or resignation of a Pope, and it remains vacant until a new Pope is selected. The selection of a new Pope is signaled to the world by white smoke rising from a chimney. If no Pope is selected at the end of a day, black smoke is used instead. Once the Pope is selected, no more smoke is used on the subsequent days.

9. Suppose the seat is vacant today. Are the following two events independent? There is black smoke tomorrow and there is white smoke in two days.

   A. Yes, independent.

   B. No, dependent.

10. Suppose the seat is vacant today. Are these two events independent? There is white smoke tomorrow and there is white smoke in two days.

    A. Yes, independent.

    B. No, dependent.

11. Observational studies are less reliable for measuring causal effects because of

    A. the possibility of confounding.

    B. the lack of any control group.

12. The inclusion of which of these kinds of control groups makes for a more trustworthy experiment?

    A. Contemporaneous control group

B. Historical control group

13. A researcher analyzes a natural experiment where, by lottery, some public students were given a free cafeteria lunch and a second group received a voucher they could use to buy food from a nearby grocery store. The first group was found to have a lower incidence of diabetes than the second group afterward. This is used as evidence showing that cafeteria food is healthy. Which of these is a better critique of that conclusion?

    A. The treatment and control groups are not comparable. Those who received the voucher likely had higher rates of diabetes to start.

    B. The analysis only describes an average effect. Some students might use the voucher to buy healthier food, but more might use it to buy junk.

Use the following information for the next three questions.

The number of hours of sleep people get in a night follows a normal curve with an average of 8 and SD = 2.

14. What percentage of people get at least 6 hours of sleep?

    A. 68%

    B. 84%

15. What percentage of people get between 4 and 12 hours of sleep?

    A. 68%

    B. 95%

16. What percentage of people get between 6 and 11 hours of sleep?

    A. 77%

    B. 82%

Use the following information for the next three questions.

A researcher wants to study the relationship between hours studied and GPA. She observes the following, where hours studied is the independent variable, and GPA is the dependent variable:

Average hours studied/week: 5	$SD_{study} = 3$	
Average GPA: 3.0	$SD_{GPA} = 0.5$	$r = 0.5$

17. What is the slope of the regression line?

    A. $\frac{1}{12}$

    B. 3

    C. 6

    D. 12

18. What is the intercept for the regression line?
    A. $\frac{31}{12}$
    B. $\frac{1}{12}$
    C. -12
    D. -6

19. Predict the average GPA of someone who studied 8 hours/week on average.
    A. 4
    B. 3.5
    C. 3.25
    D. 3

20. Suppose another regression predicts GPA from time spent taking private tennis lessons, using observational data. All assumptions for regression analysis are met and an hour of tennis predicts a bigger improvement in GPA than does an hour of studying. What is an appropriate conclusion?
    A. Someone who wants a good GPA should quit studying and spend more time playing tennis.
    B. The conclusion depends on if all subjects were observed contemporaneously.
    C. The study cannot be trusted because it was not double blind.
    D. This is likely spurious correlation because students who take private tennis lessons might also have access to private tutors.

Use the following information for the next three questions.

The table below summarizes the endowments for three universities. Suppose no other universities exist and 2023 was the first year in existence for Napier University. For 2021 and 2022, there are only two values (do not include Napier as a zero).

University	2021	2022	2023
Columbia	5	5	5
Baylor	1	3	3
Napier	-	-	7

Table 1: University Endowment Values (in billions)

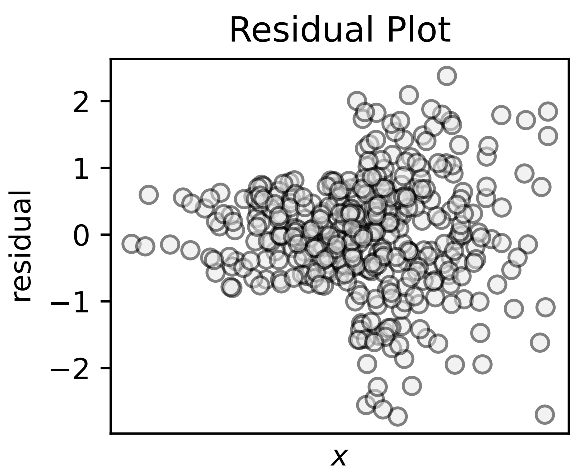
21. Considering only the 2021 data, what is Baylor's endowment in standard units?
    A. -2
    B. -1
    C. 0

D. 1

22. Considering only the 2022 data, what is Baylor's endowment in standard units?
    A. -2
    B. -1
    C. 0
    D. 1

23. Considering only the 2023 data, what is Baylor's endowment in standard units?
    A. $z = -2$
    B. $-2 < z < -1$
    C. $z = 1$
    D. $-1 < z \leq 0$

24. The following residual plot is _____ and a possible explanation is _____.



    A. Homoscedastic; x comes from a skewed distribution.
    B. Heteroscedastic; y is easier to predict at high x values.
    C. Heteroscedastic; a researcher sorted the data by x value and dropped observations with extreme residuals until the RMS error was sufficiently low.
    D. Heteroscedastic; a researcher sorted the data by x value and dropped observations with extreme residuals until the RMS error was sufficiently high.

Use the following information for the next four questions.

A scientist collects data and runs a regression, predicting the measured values y_{meas} from x. However, due to measurement error, the measured data is actually a distortion of the true data, $y_{\text{meas}} = 1 - y_{\text{true}}$.

25. How does the correlation between x and the true y data relate to the correlation between x and the measured data?

    A. $r_{\text{meas}} = r_{\text{true}}$

    B. $r_{\text{meas}} = -r_{\text{true}}$

    C. $r_{\text{meas}} > r_{\text{true}}$

    D. Not enough information

26. How does $\text{SD}_{y_{\text{true}}}$ relate to $\text{SD}_{y_{\text{meas}}}$?

    A. $\text{SD}_{\text{meas}} = \text{SD}_{\text{true}}$

    B. $\text{SD}_{\text{meas}} = -\text{SD}_{\text{true}}$

    C. $\text{SD}_{\text{meas}} > \text{SD}_{\text{true}}$

    D. $\text{SD}_{\text{meas}} < \text{SD}_{\text{true}}$

27. How does the slope, m, of the true regression line compare to the regression line for the measured data?

    A. $m_{\text{meas}} = m_{\text{true}}$

    B. $m_{\text{meas}} = -m_{\text{true}}$

    C. $m_{\text{meas}} > m_{\text{true}}$

    D. $m_{\text{meas}} < m_{\text{true}}$

28. If $\hat{y}_{\text{meas}} = 3x$ is the regression equation for the measured data, what is the y-intercept in the regression equation for the true data?

    A. -1

    B. 0

    C. 1

    D. 3

Short Answer

Show your work and justify your answers for full credit.

29. A maverick statistician invents a new measure of spread, called *Median Spreadification* (MS). It is calculated by substituting the median for the average in the SD formula.

$$\text{MS} = \sqrt{\frac{1}{n}\left((x_1 - \text{median})^2 + \cdots + (x_n - \text{median})^2\right)}$$

    (a) (2 points) If data follows a normal curve, will 68% of the data fall within one MS of the median?

(b) (4 points) The statistician discovers that MS $\geq$ SD for any data set. Suppose a data set is skewed with mostly low values but there is a very long right tail. Use a graphical argument to show, for a skewed enough data set, more of the data will fall within one MS of the median than will fall within one SD from the average.

30. Rosa is accused of a crime. Everyone agrees that she did it, but many find the law unjust. She is judged by three random jurors, each one of two possible types: nullifiers and rule-followers. A nullifier will vote that Rosa is innocent. The rule-followers will vote that Rosa is guilty. The two types are equally likely, drawn with replacement from the box below. Suppose a guilty verdict requires all three jurors to vote guilty.

nullifier	rule-follower

(a) (3 points) What is the probability that the jury reaches a unanimous verdict, either innocent or guilty?

(b) (3 points) What is the probability that Rosa is judged guilty?

(c) (4 points) Now, suppose the jury of three is randomly drawn from a pool of ten potential jurors. It is known that there are five of each type. The draws can be done with replacement or without replacement. If a juror is selected multiple times, they simply get multiple votes (counting as multiple people). Should Rosa's defense lawyer ask for the jury to be drawn with replacement or without replacement?